When the Earth Shakes

Understanding Global Earthquake Patterns · 1995-2023

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Presentation Goal

Our goal is to find out whether magnitude alone explains earthquake impact, or whether depth, region, alert level, intensity, and tsunami status provide a better picture of earthquake severity.

Contents

1. EARTHQUAKE METRICS
2. SEISMIC BEHAVIOR
3. HISTORICAL PATTERNS
4. AFTERSHOCK
5. CONCLUDING INFORMATION

The Epicenter of it all

Major earthquakes cause widespread destruction - Understanding the data saves lives.

31
red alert events
highest danger level

629
tsunami-triggering
events recorded

M 9.1
strongest recorded
Indian Ocean, 2004
Tohoku, Japan, 2011

29 yr
of seismic data
1995-2023

Key Insight:

The deadliest earthquakes cluster around “The Ring of Fire”, and the most impactful earthquakes occur above a 200 km depth.

How are Earthquakes Measured

ALERT LEVELS

Estimates Severity

KEY METRICS

Magnitude
Energy released at the source (Log Scale)

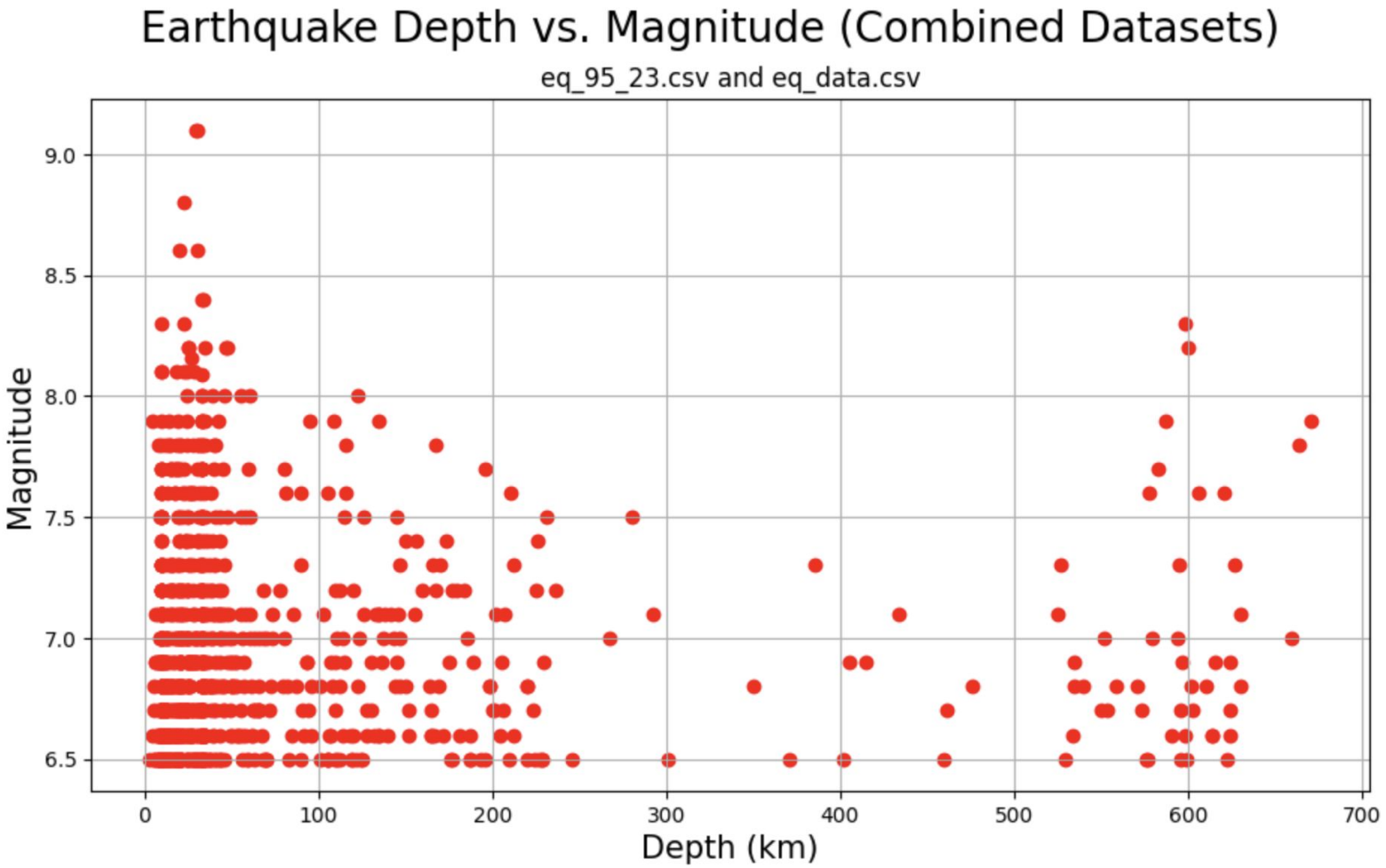
CDI - Community Felt Intensity
How strongly people reported feeling the quake

MMI - Modified Mercalli Intensity
Instrumental Ground Shaking

Sig - Significance
A number (typically 0–1000) that describes how "significant" the event is based on a combination of magnitude, PAGER alert level, maximum MMI, and the number of felt reports. Higher numbers indicate a more impactful event.

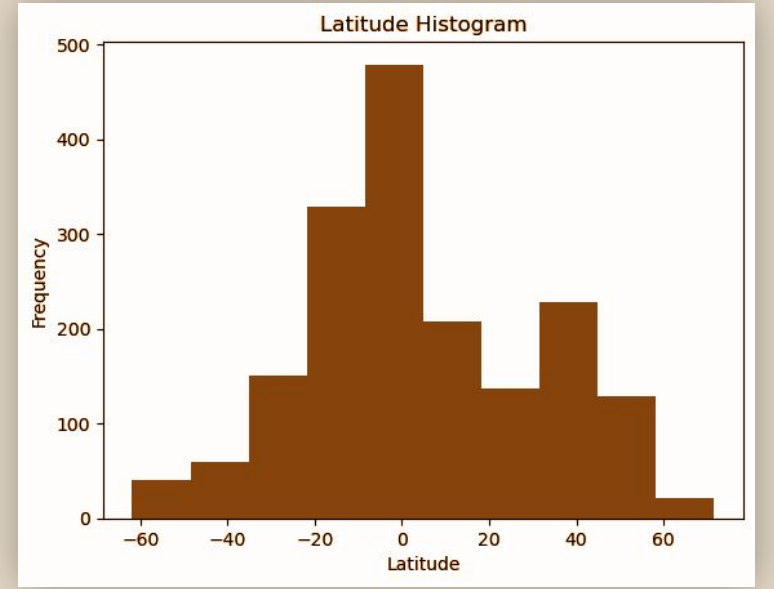
Depth Matters:

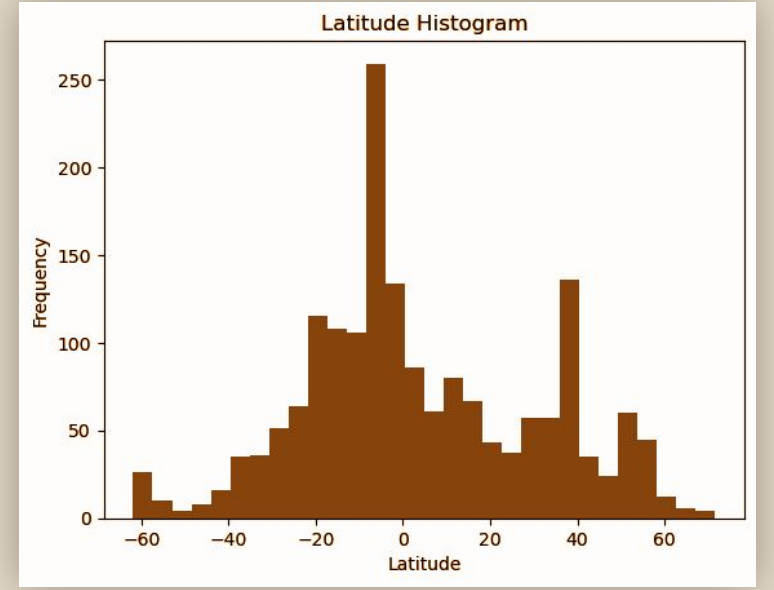
Shallow quakes (<70km) cause far mores surface damage than deep ones. Avg: 75km Range: 2.7 - 670km


Earthquake Depth vs. Magnitude (Combined Datasets)
eq_95_23.csv and eq_data.csv
Magnitude
9.0
8.5
8.0
7.5
7.0
6.5
0
100
200
300
400
500
600
700
Depth (km)

Earthquakes by Latitude

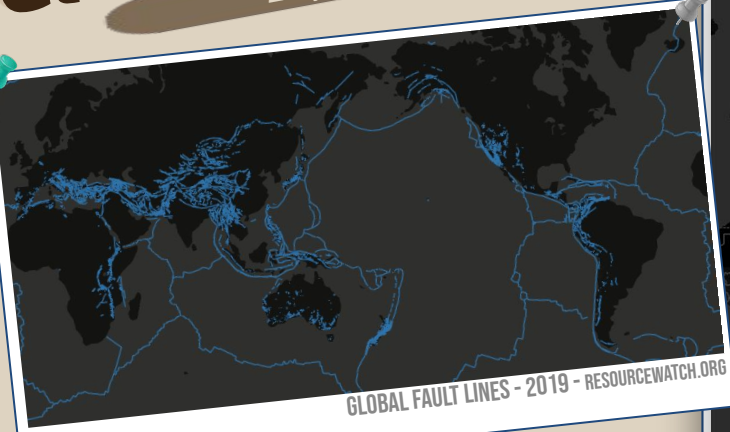
Where in the world?

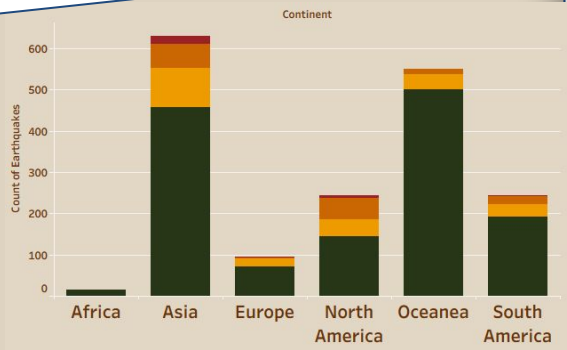


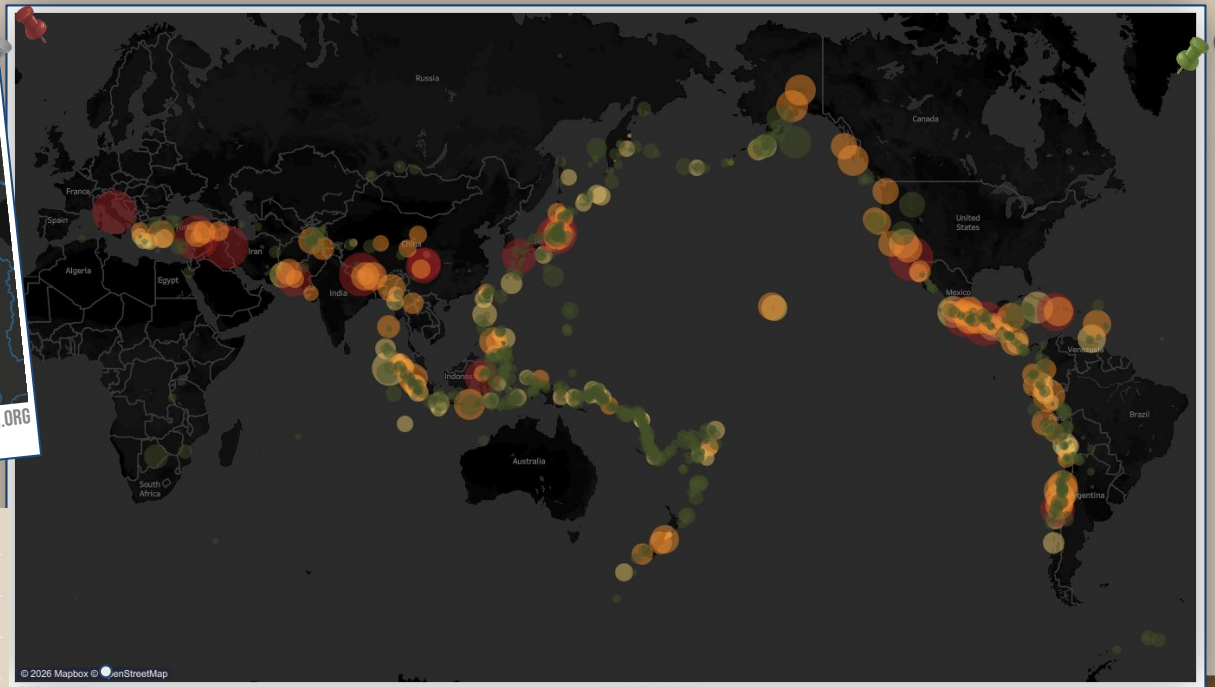


Global Earthquake Distribution

29 years of earthquakes, their sig levels, and alerts







SIG LEVEL

2910 2500 2000 1500 1000 650

ALERT LEVEL

GREEN YELLOW ORANGE RED

Visualization in Python

An eternity of coding… Explanation of visualization technique

```python
# Create pivot table
levels = merged_data.groupby(['continent', 'alert']).size().reset_index(name='count')
# Only use colors for columns that exist in data
fig = px.bar(levels,
             x='continent',
             y='count',
             color='alert',
             barmode='stack',
             color_discrete_map={
                 'green': 'darkolivegreen',
                 'yellow': 'goldenrod',
                 'orange': 'chocolate',
                 'red': 'firebrick'
             },
             # Organize the order each alert is displayed
             category_orders={'alert': ['green', 'yellow', 'orange', 'red']},
             title='Earthquake Alert Levels by Region')

# Adjust the plot size
fig.update_layout(width=1200, height=600,
                  plot_bgcolor = 'white')

fig.show()
```

REFERENCE PLOTLY.COM - HTTPS://PLOTLY.COM/PYTHON/PLOTLY-EXPRESS/

Grouped data by continent and alert

Created a stacked bar chart of earthquake counts sectioned by continent and alert levels

Mapped colors to alert levels and designated color display order

Mapped alert colors

Map scatterplot visual using significance as dot size and alert levels for color

Modified background color and dot opacity

```python
# Color mapping for alert levels
color_map = {
    'red': 'firebrick',
    'orange': 'chocolate',
    'yellow': 'goldenrod',
    'green': 'darkolivegreen'
}

# Create a scattermap of earthsuake significance represented by dot size and color represented by alert
fig = px.scatter_map(
    merged_data,
    lat='latitude',
    lon='longitude',
    # Base size by significance level
    size='sig',
    # Base color by alert level
    color='alert',
    # Assign colors to alerts
    color_discrete_map=color_map,
    #Set the title of the hover box
    hover_name='title',
    #Data to show when hovering over a dot
    hover_data={'magnitude': True, 'depth': True, 'sig': True, 'alert': True},
    size_max=30,
    zoom=1,
    title='Earthquake Significance Map',
    map_style='carto-darkmatter'
)
# Modify the background color
fig.update_layout(
    paper_bgcolor='#111',
    font_color='white',
    title_font_size=18,
    legend_title_text='Alert Level'
)
# Adjust the opacity of the dots
fig.update_traces(marker=dict(opacity=0.2))
fig.show()
```

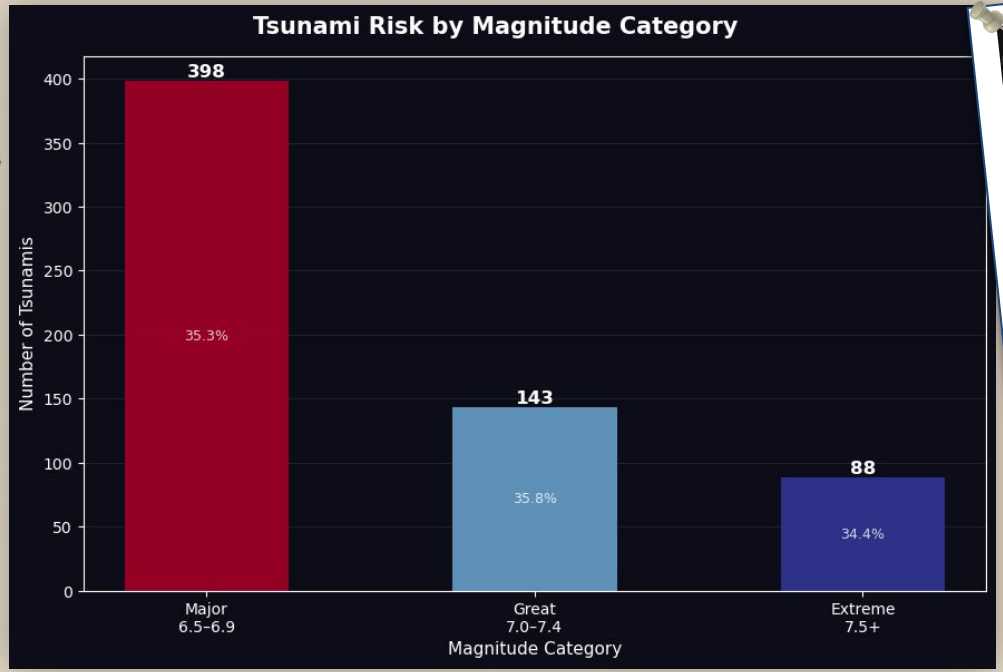
REFERENCE
PLOTLY.COM - HTTPS://PLOTLY.COM/PYTHON/SCATTER-PLOTS-ON-MAPS/

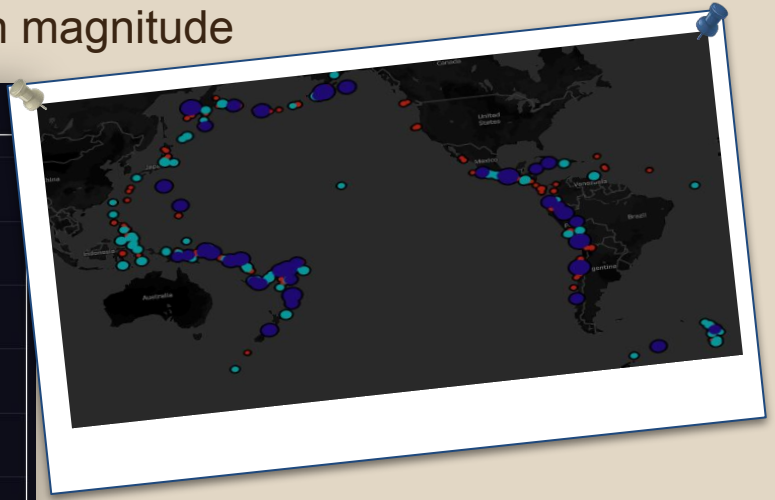
HEADACHE LEVEL

NONE | MODERATE | BAD | SEVERE

Tsunami risk

Visualizations on magnitude

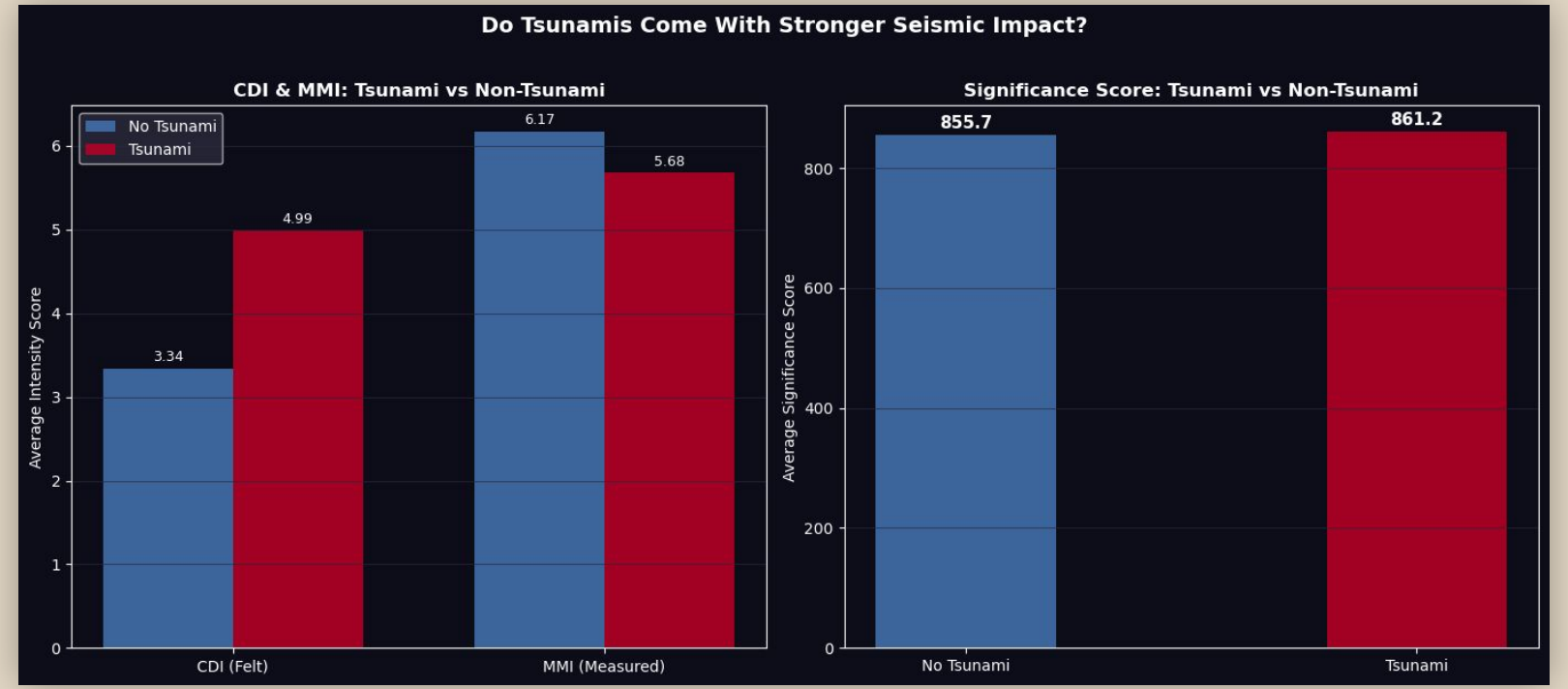




- 54% of earthquakes cause a tsunamis

Tsunami risk

Seismic Impact



Seismic Events

The most powerful earthquakes in the data set by significance

2004
9.1 MAG
Sumatra, Indonesia (Ocean)
Massive Tsunami,
280,000+ deaths
1.1 mil displaced

2005
8.6 MAG
Sumatra, Indonesia
Small Tsunami,
915+ deaths,
$60+ mil in humanitarian aid
form Australia

2010
8.8 MAG
Maule, Chile
525 deaths
Multiple Tsunamis,
21% buildings damaged

2011
9.1 MAG
Tohoku, Japan (Ocean)
Massive Tsunami,
19,700+ deaths,
$360 billion in damage

2012
8.6 MAG
Indonesia, Indian Ocean
31 inch Tsunami,
10 died of shock,
minor damage, bridge collapse

Information Impact

- **Depth:** Earthquakes occur more at depths shallower than 200km and have wider severity ranges
- **Frequency:** Earthquake occurrences happen closer to the equator
- **Alerts:** A large majority of earthquakes have green alert levels, but Asia and North America experience higher alert levels than most
  - Asia and Oceania Experience the most earthquakes.
- **Tsunamis:** 54% of earthquakes cause a tsunami.

Thank you

We hope this news was truly groundbreaking for you.